GOOGLE CLOUD AI BUILDER CUP 2026 · SUSTAINABILITY & SOCIAL IMPACT

SafeGround

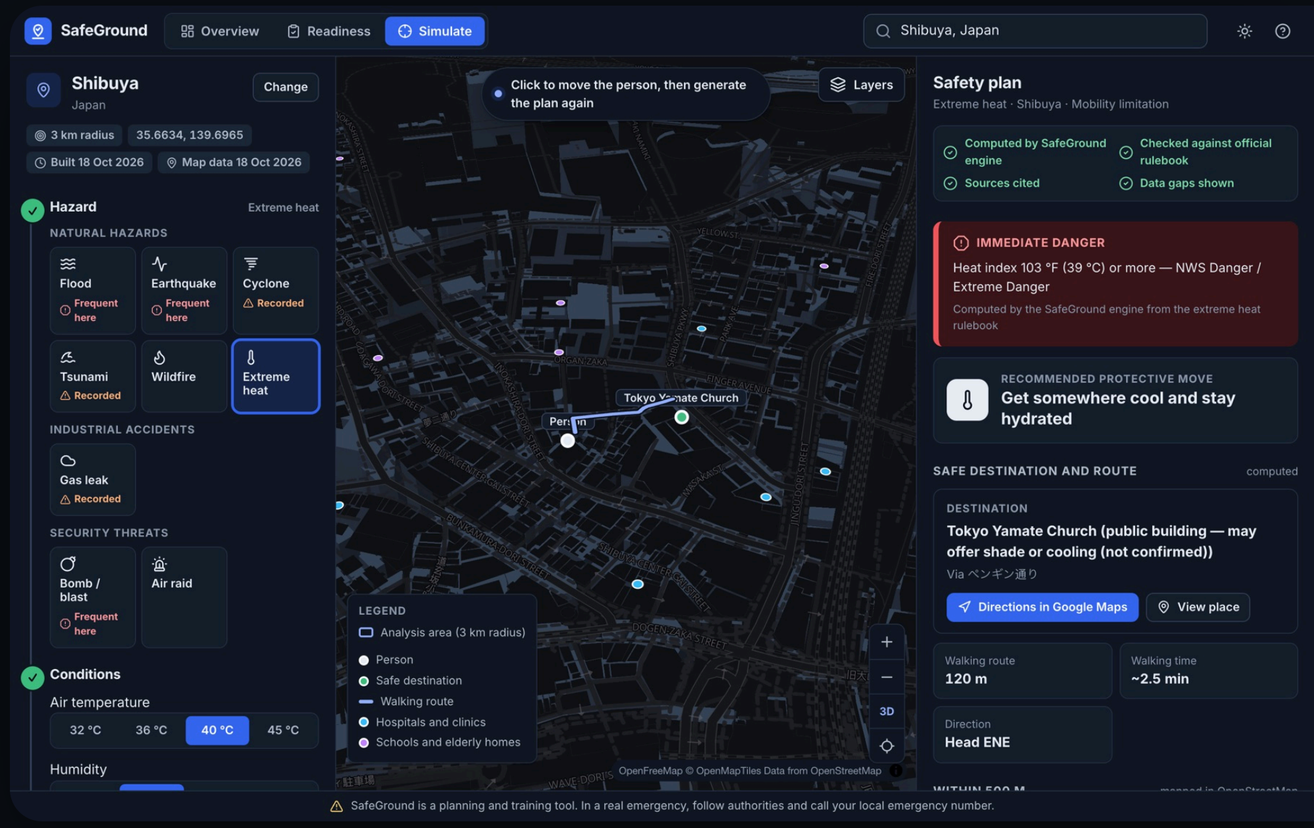
Hazard readiness and response planning for any place on Earth.

What hits a place, how ready it is, and what each person should do if it strikes right here — with a route to safety and every claim traced to its source.

9 hazards · official guidance · Google Earth Engine data

Gemini agents on Google ADK · Cloud Run

Live: safeground-653135417611.asia-south1.run.app



THE PROBLEM

The right move depends on the hazard and the exact spot — generic advice fails people

- **Disasters hit specific places:** Paiporta’s flash flood (2024), the Beirut port explosion (2020), the Antakya earthquake (2023), the Lahaina wildfire (2023).
- **Planners can’t see it in one place:** which hazards are on record, which blocks lack a refuge, a hospital or a shelter, and who needs help first.
- **A chatbot alone guesses:** in our test, Gemini without computed facts gave the wrong danger level in **14 of 20** cases.

ONE STREET CORNER IN SHIBUYA, TOKYO · FOUR HAZARDS · FOUR RIGHT MOVES

Earthquake

Drop, cover, hold

When shaking stops and a building is damaged: open area 185 m away.

Air-raid alert

Go down

Underground car park 120 m away (1.6 min walk).

Chlorine leak

Stay in & seal

Never go down — gas collects low. Exits crosswind, never downwind.

Tsunami warning

Stay put

This spot is above 30 m (Ready.gov’s 100 ft) — follow officials.

Shibuya moves computed by SafeGround’s engine (35.6612 N, 139.6985 E) · eval/RESULTS.md

THE SOLUTION

One platform, three steps — for any place on Earth

1 · Overview

What hits this place?

- Hazard profile from the place's **own records**: floods, rain extremes, earthquakes, cyclones, fires, checked news.
- Exposure: residents, low ground, schools, elderly homes, hospitals, shelters, open areas.
- Country emergency numbers.

2 · Readiness

How ready is it?

- An AI investigator picks the checks that matter here.
- Readiness **before / during / after** for **people, animals and nature**.
- Ranked, map-linked actions; every number traced to a dataset.

3 · Simulate

If it happens here, what now?

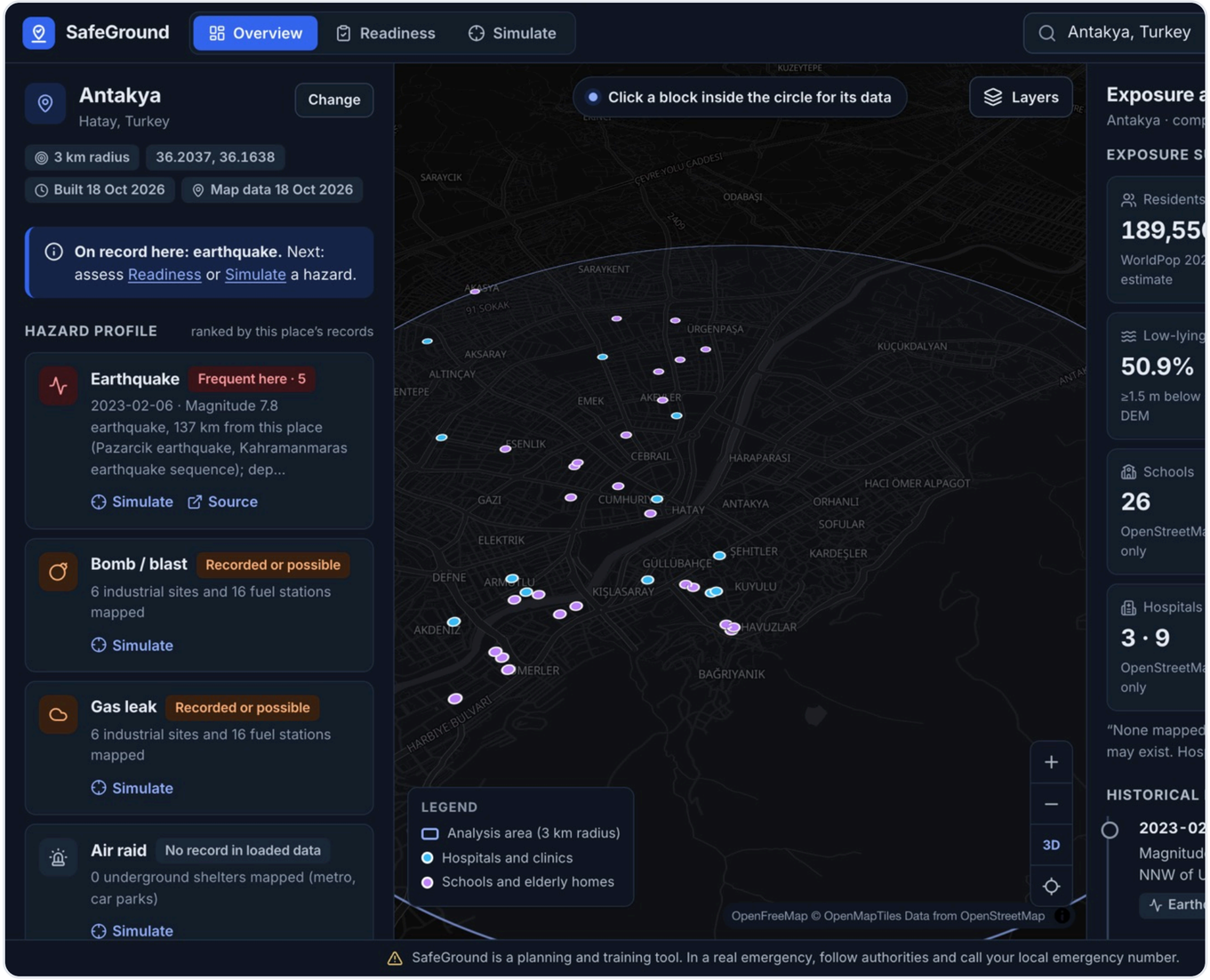
- **9 hazards**, conditions you set, a spot and a person's situation.
- Danger zone from official guidance → safe move → nearest safe place → route + **Google Maps directions**.
- Plain-English plan; every step cites official advice.

For: city and district disaster teams · companies protecting sites and staff · schools and hospitals planning drills · NGOs pre-positioning relief · residents.

A planning and training tool: no alerts are sent, no personal data is collected, and nothing replaces official instructions.

A hazard profile built from the place’s own records

- **Earthquake — frequent here:** 5 strong earthquakes on record from USGS, including M7.8 on 2023-02-06.
- Hazards are sorted by these records, and the simulator uses the same order — **proactive, not a fixed menu**.
- **Exposure** from Google Earth Engine and OpenStreetMap: residents (WorldPop), height above sea level (Copernicus DEM), low-lying blocks, tree cover 2016→2025 (Dynamic World), schools, elderly homes, hospitals, underground shelters, open areas.
- Click any block on the map for its population, height and land cover.



History: USGS earthquake catalogue · GDACS · Global Flood Database · CHIRPS / ERA5-Land · FIRMS · link-checked news

Context-based readiness, not a fixed checklist

1 · Past disasters

Gemini + Google Search grounding; every link is opened and must mention the place and the year. The 2020 explosion page blocks bots, so it was checked on its Internet Archive copy.

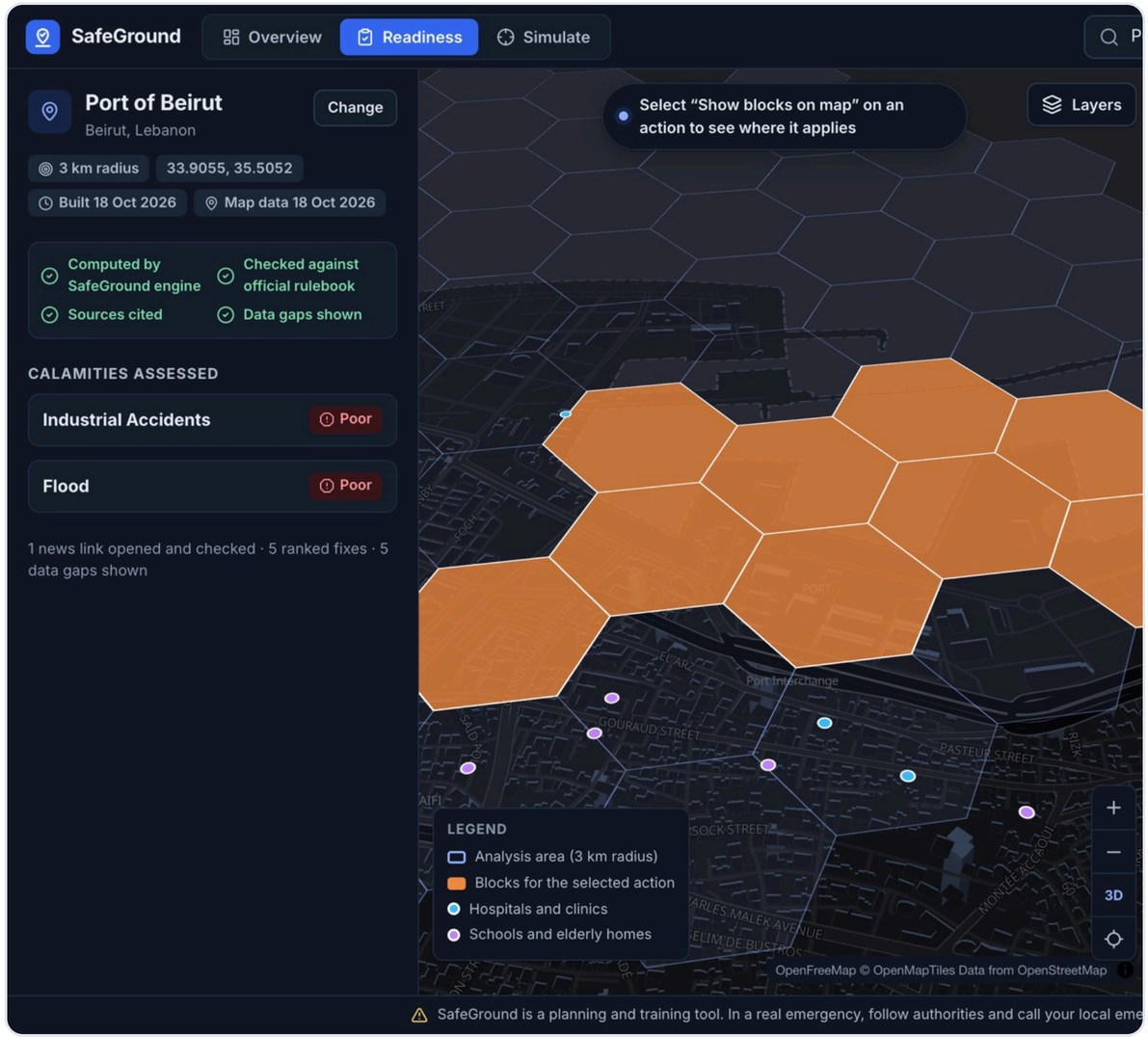
2 · Investigator agent

Chooses the tools for *this* place: flood exposure, refuge / hospital / fire-station walking access, industrial exposure, land-cover change, animal facilities.

3 · Writer → validator

Every number must appear in a tool result; links must come from the checked list.

Found: 6,074 residents in flood-prone blocks lack a nearby vertical refuge; 18 industrial sites within 4.2 km. **Lahaina:** no hospital mapped within 3 km; 0% of residents within a 1 km walk of a fire station; bare land 1.8% → 15.4% (2016 → 2025).



3 · SIMULATE · 9 HAZARDS, ONE ENGINE

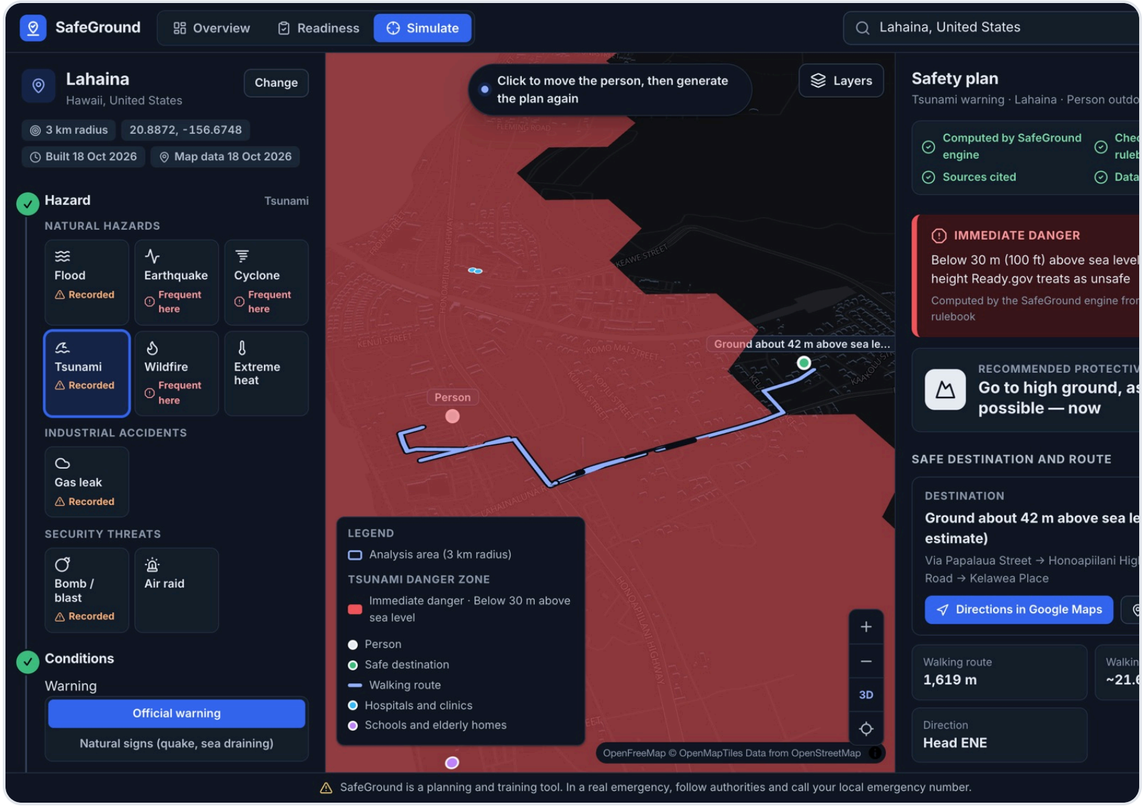
Official guidance → danger zone → safe move → route

HAZARD	DANGER COMPUTED FROM	SAFE MOVE AND TARGET	GUIDANCE
Flood	Low land (DEM), surface water, rain; reported flooding overrides	Go up: tall building / high ground, no underpasses	State flood do’s and don’ts
Earthquake	Shaking now / after shaking	Drop, cover, hold; then nearest open area	Ready.gov (FEMA)
Cyclone	Surge-exposed vs wind-only blocks; Saffir-Simpson	Go up, or interior room	Ready.gov, NOAA NHC
Tsunami	Height vs Ready.gov’s 100 ft (30 m)	Nearest ground ≥ 30 m, inland	Ready.gov
Wildfire	Fire + wind: spread ≈ 10% of wind speed	Leave early, crosswind, never downwind	Ready.gov; Cruz & Alexander 2019
Extreme heat	NWS heat index (temperature, humidity)	Get cool; nearest public building	Ready.gov, US NWS
Gas leak	ERG 2024 distances, shaped by wind	Shelter and seal, or move crosswind	ERG 2024, Ready.gov
Bomb / blast	DHS/FBI stand-off distances	Evacuate, or shelter away from glass	DHS/FBI, ERG 2024
Air raid	Whole-area alert	Nearest underground shelter	Civil-defence drill guidance

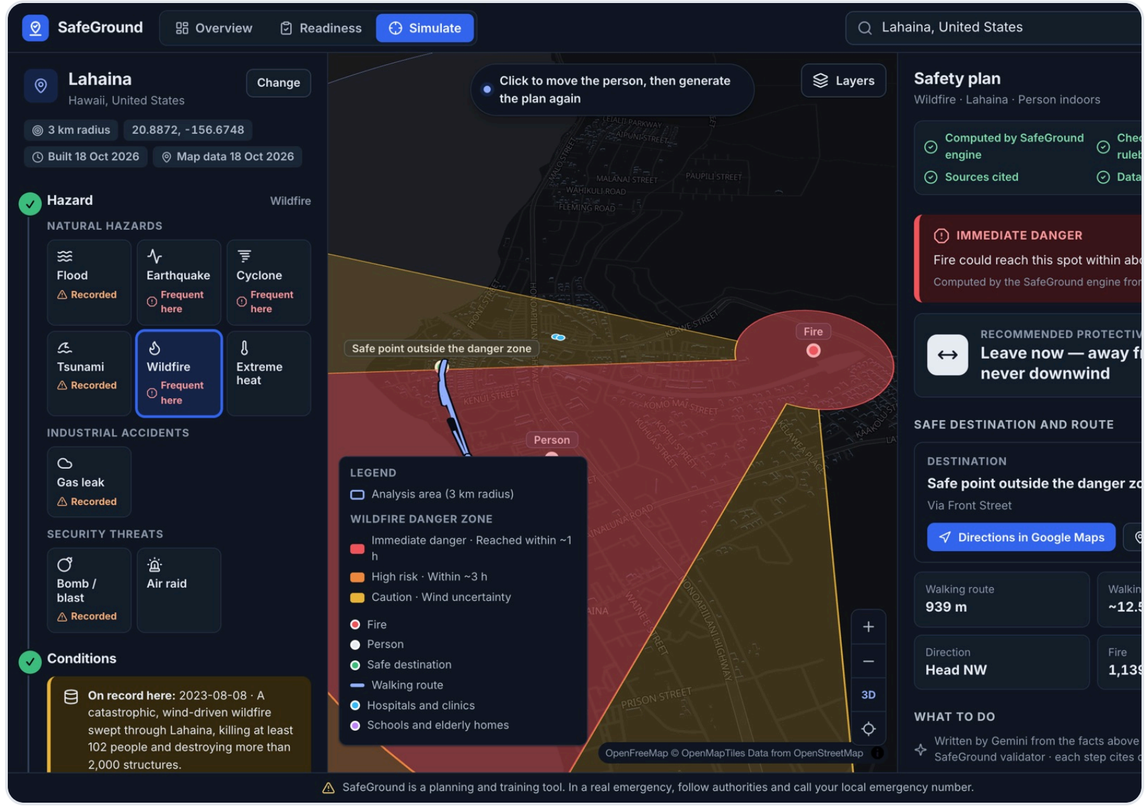
Each hazard is a rulebook file (zone model, conditions, allowed / forbidden moves, verbatim advice with sources). The UI builds its controls from these files. Every plan also lists schools, elderly homes and clinics within 500 m.

3 · SIMULATE · LAHAINA, HAWAII

A plan a person can follow — with directions in Google Maps



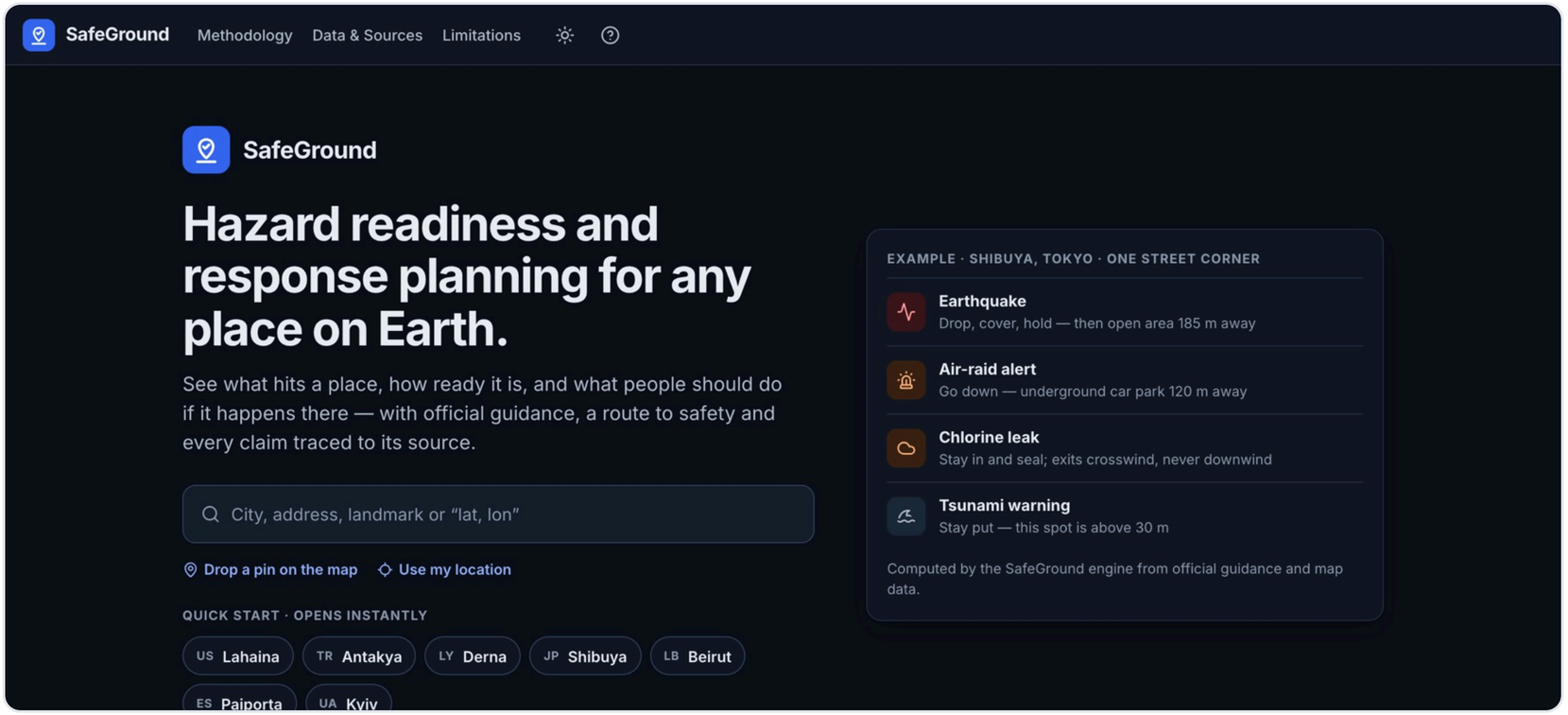
Tsunami warning: this block is below 30 m, so the plan says go to high ground — ground ~42 m above sea level, 1.6 km ENE, a 22-min walk that avoids underpasses.

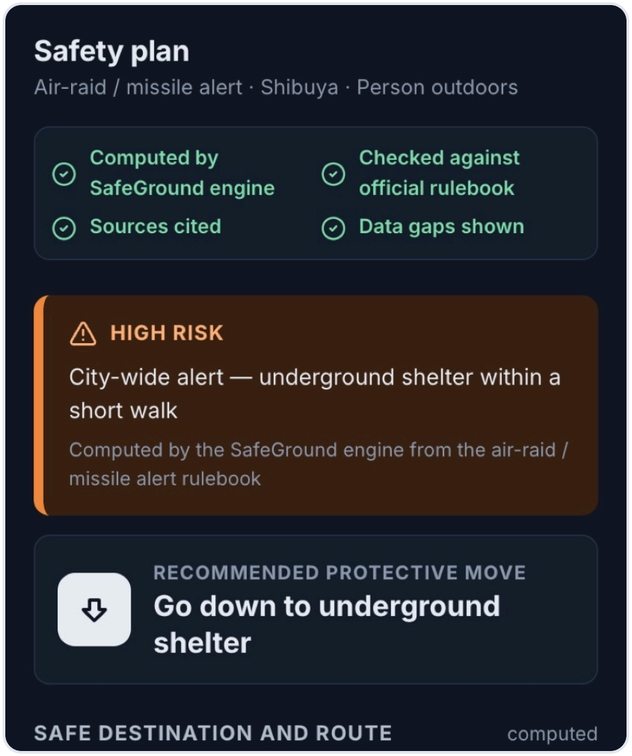


Wildfire, wind from NE at 6 m/s: the fire could reach this spot within an hour. Leave now — 939 m NW, out of the spread zone, never downwind.

ANY PLACE ON EARTH

Search, drop a pin or pick a quick start — the place is built live





1 · Map data
OpenStreetMap (Overpass); OpenFreeMap vector tiles when it is busy

→

2 · Earth Engine
Population, buildings, height, water, land cover for ~300 blocks

→

3 · History
Flood DB, rain extremes, fires, USGS quakes, GDACS alerts, checked news

→

4 · Same agents
Readiness report and safety plans, validated by code

Ready now: Paiporta (ES) · Shibuya (JP) · Port of Beirut (LB) · Antakya (TR) · Kyiv (UA) · Lahaina (US) · Derna (LY) — any other place in ~2 minutes

BUILT ON GOOGLE CLOUD & GOOGLE AI

Code computes the facts. Gemini reasons and explains.

DATA

Google Earth Engine (live)
WorldPop · Open Buildings · Dynamic World 2016→2025 · Copernicus DEM · JRC water · WDPA

Earth Engine history
Global Flood Database · CHIRPS / ERA5-Land rain · FIRMS fires

Open data
OpenStreetMap walking graph and facilities · USGS · GDACS · Photon geocoder

Official rulebooks (9)
Ready.gov, ERG 2024, DHS/FBI, NOAA NHC, NWS, flood and civil-defence guidance

AGENTS (GOOGLE ADK · GEMINI 3.5 FLASH)

NewsResearcher
Google Search grounding; every link opened and checked (Internet Archive fallback)

Readiness investigator → writer
Tool-calling agent picks checks; writer outputs a typed ReadinessReport

SafetyPlanner
Typed SafetyPlan from computed facts and rulebook lines only

Validator (code)
Rejects a wrong tier, forbidden moves, uncited advice, numbers not in the facts, unsafe phrases → feedback loop (3 tries) → official-text fallback

SERVING

Cloud Run
FastAPI + engine, scales to zero, dedicated service account

Cloud Build · Secret Manager
Source deploy; Gemini key never in code

Engine (code)
9 hazard zones, Dijkstra routes that avoid danger and underpasses, access fields, overview

Web
Landing + map-first workspace; provenance badges; print / PDF / share; WCAG 2.2 AA; dark-first

Repo: safeground/ (engine, agents, api, rulebooks) · web/ · eval/ · tests/ (22 tests)

ACCURACY & TRUST

Every answer is checked by code before anyone sees it

Validator catches invention

In one readiness report the writer’s first two drafts added up residents on their own. Both were rejected as numbers not in any tool result; the third draft passed.

Sources are opened, not trusted

Each news link is fetched and must mention the place and the year. Sites that block bots are checked on their Internet Archive copy. Unchecked links never reach a report.

Unsafe moves are blocked

Basement in a gas leak, downwind in a wildfire, the shore in a tsunami, a doorway in an earthquake, wading through flood water — rejected by rule.

Labelled estimates

Flood, surge and tsunami tiers, gas and fire zones and population are shown with their method and limits.

Unknown stays unknown

Hospital capacity, basements, distance from the coast and whether a building is open are shown as gaps.

No guessed numbers

Emergency numbers come from a checked table per country; if a country is missing, the app says so.

EVALUATION · 20 SAFETY-PLAN CASES

Same Gemini model: with SafeGround’s facts vs. on its own

CHECK	SAFEGROUND	GEMINI ALONE
Danger level matches the official-distance tier	100%	30%
Main move allowed for that tier	100%	70%
Plans with invented distances or times	0	3
Plans with an unsafe instruction	0	0

Cases at real street locations: a night ammonia leak with wind, bomb threats, an air-raid alert and very heavy rain with reported waterlogging, across 6 situations. Scoring is rule-based against the engine’s official zone; 19/20 SafeGround plans passed the validator the first time. The 5 newer hazards (earthquake, cyclone, tsunami, wildfire, heat) passed validation on the first attempt in live tests and are covered by unit tests; they are not yet in this comparison.

IMPACT

Findings a disaster team can act on — on four continents

Antakya · earthquake, flood

42,007

residents in flood-prone blocks with no flood refuge within a 300 m walk

Lahaina · wildfire

0

hospitals mapped within 3 km; 0% of residents within a 1 km walk of a fire station; bare land 1.8% → 15.4% since 2016

Port of Beirut · coast

182 / 264

blocks below 30 m above sea level; 18 industrial sites within 4.2 km

Shibuya · crowds

11 + 2

schools and elderly homes within 500 m of one station corner — who to help first

Social impact: drills, refuges and evacuation routes planned from a place's real records, for the people least able to move — elderly homes, schools, hospitals.

Sustainability: readiness covers animals and nature too — tree and wetland loss (Dynamic World 2016→2025), vet access, protected areas.

Numbers from validated readiness reports and the place overview (data/cache/)

LIMITS & NEXT STEPS

What it is not — yet

Limitations

- Flood, surge and tsunami tiers are estimates from a 30 m elevation model; distance from the coast is not measured.
- ERG distances are made for transport incidents; the wildfire spread is a rule of thumb, not a fire model.
- OpenStreetMap gaps: hospital capacity, basements, elderly homes, animals.
- Each place covers a 3 km circle; conditions (wind, rain, heat) are set by the user.
- Official advice is from US and Indian sources; local guidance may differ.

Next

- Live conditions and alerts (Google Weather API, Flood Hub, GDACS) to keep plans current.
- Local official guidance and languages per country; voice for people who can’t read the screen.
- Area impact briefs with actions by role (residents, responders, city) and drill exports.
- Team workspaces and saved scenarios for organisations.
- Human review of readiness reports with local disaster-management officers.

Try it: safeground-653135417611.asia-south1.run.app · In a real emergency, follow official instructions and call the local emergency number.